M1R Random Matrices

Sheehan Olver

May 13, 2026

Contents

In these notes we introduce random matrices, introducing some basic theory combined with practical experiments. Much of the theory goes beyond the typical mathematical knowledge of 1st year students so these notes are really just the tip-of-the-iceberg: random matrices have a very rich theory combining aspects of probability, statistics, combinatorics, group theory, approximation theory, functional analysis, complex analysis, integrable systems, statistical mechanics, and numerical analysis.

We divide the notes into four chapters:

I. *Central Limit Theorem.* Before delving into random matrices we begin with a scalar version of universality: the Central Limit Theorem (CLT). This theorem says when we add independent, identically distributed (iid) random variables whose moments are finite the behaviour of the sum tends to that of a normal distribution. We sketch the proof of this result using the Method of Moments, a technique that is adaptable to studying random matrices.

II. *Universality.* Eigenvalue statistics have universality properties that emerge independent of the details of the distribution of their entries. For example, if you plot a histrogram of the eigenvalues of a large symmetric matrix with iid entries whose moments are finite it will always look like a semicircle (the *semicircle law*). We sketch the proof of this result again using the Method of Moments and discuss experimentally other statistics like the largest eigenvalue.

III. *Free Probability.* Algebraic manipulations of random matrices have the property that their eigenvalue distribution can be deduced via *Free Probability*, which can be viewed as a non-commutative version of probability that originally arose in operator algebra theory. This leads to a non-commutative analogue of convolution. We observe this phenomena experimentally and discuss some of the beautiful properties of the *free convolution*.

IV. *Randomised Linear Algebra.* We can incorporate randomness into linear problems like least squares (i.e., regression) and achieve fairly accurate results with much less computational cost. This chapter introduces least squares and explores, experimentally, the practical realisation of randomised linear algebra.

Chapter I

Central Limit Theorem

Before delving into universality for random matrices we explore the scalar random variable version: the Central Limit Theorem (CLT). You would have already seen the *Normal distribution* $\mathcal{N}$ (with mean zero and variance one), which has a Probability Density Function (PDF) which is a Gaussian:

$$\frac{\mathrm{e}^{-x^2/2}}{\sqrt{2\pi}}.$$

The CLT is the fundamental result that when we add many independent samples of a random number it behaves like a Normal distribution. More precisely, if $X_1, \ldots, X_n$ are all iid and (for simplicity) have zero mean and variance:

$$\mathbb{E}X_k = 0, \qquad \mathbb{E}X_k^2 = 1,$$

then we claim that the rescaled sum

$$S_n := \frac{X_1 + \cdots + X_n}{\sqrt{n}}$$

looks like $\mathcal{N}$ as $n \to \infty$.

We will show this using the Method of Moments: we want to show the moments $\mathbb{E}S_n^k$ tend to the moments of $\mathcal{N}$. When moments do not grow too fast, convergence in moments tells us convergence in probability densities (the precise details of this are beyond the scope of these notes), so it suffices to show convergence of moments to establish the CLT. This argument serves as a blueprint for a similar Method of Moments that can be used to prove universality for random matrices.

I.1 Experimental observation of the Central Limit Theorem

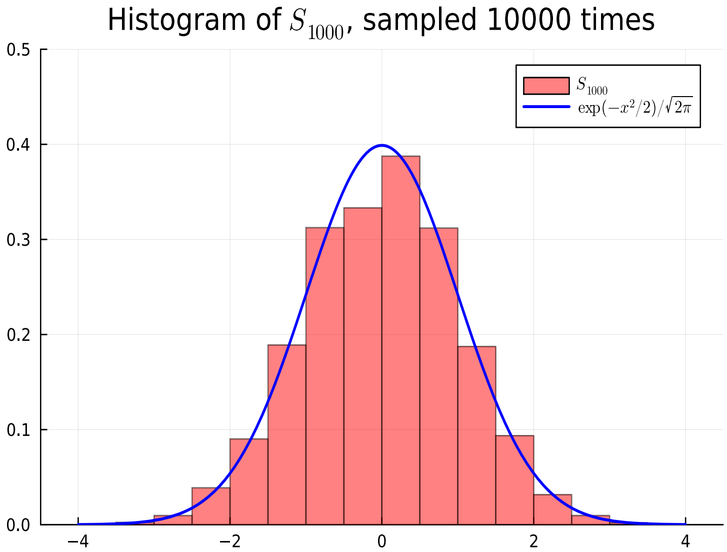
Consider sampling iid discrete random variables $X_1, \ldots, X_n$ each taking values ± 1 with equal probability, so that their expectation is zero and variance is one:

$$\mathbb{E}X_k = (-1)\frac{1}{2} + 1\frac{1}{2} = 0, \qquad \mathbb{E}X_k^2 = (-1)^2\frac{1}{2} + 1^2\frac{1}{2} = 1.$$

By summing and rescaling these variables we can get a new random variable S_n:

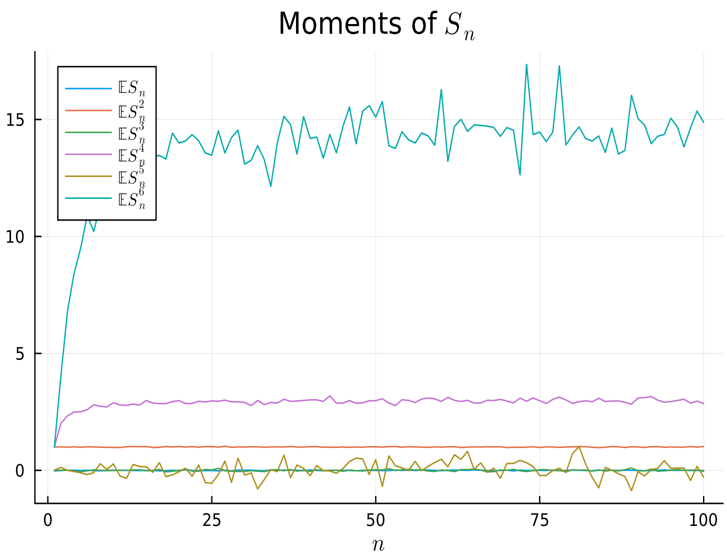
$$S_n := \frac{X_1 + \cdots + X_n}{\sqrt{n}}.$$

For large n we can compare a histogram of samples of S_n to a Gaussian:



They look very close! In fact, by making n and the number of samples large enough we can make the histogram look arbitrarily close to a Gaussian. This phenomena is independent of the distribution of X_k provided the moments are finite: we always get a normal distribution (shifted and scaled to match the mean and variance).

We are going to show why this phenomena is true using the Method of Moments: we want to show the moment $\mathbb{E}S_n^k$ tend to the moments of a Normal distribution. We can deduce the moments experimentally by averaging over many samples and letting n increase:



A sensible conjecture then is that $\mathbb{E}S_n^k$ is zero when k is odd whilst

$$\mathbb{E}S_n^2 \to 1, \quad \mathbb{E}S_n^4 \to 3, \quad \mathbb{E}S_n^6 \to 15.$$

Thus we are left with the tasks of: (1) deducing an exact formula for these moments, and (2) proving that the Gaussian has the same moments.

I.2 Moments of the normal distribution

We will use the Method of Moments, so our first step is to establish an explicit formula for the moments of a Normal distribution, i.e., we want to compute

$$\mu_k^N := \int_{-\infty}^{\infty} x^k \frac{\mathrm{e}^{-x^2/2}}{\sqrt{2\pi}} \mathrm{d}x.$$

We will relate these integrals to the following combinatorial object:

Definition 1 (double factorial)**.**

$$(2m-1)!! := \frac{(2m)!}{2^m m!} = 1 \cdot 3 \cdot 5 \cdots (2m-1).$$

Note this satisfies a simple recurrence:

$$(2m-1)!! = (2m-1) \cdot (2m-3)!! \quad \text{for} \quad m > 1.$$

We can check the first few double factorials:

$$1!! = 1, \quad 3!! = 3, \quad 5!! = 15$$

which exactly match our predicted values for the first few even moments $\mathbb{E}S_n^2, \mathbb{E}S_n^4, \mathbb{E}S_n^6$. We claim this holds true for all even moments: $\mathbb{E}S_n^{2m} = (2m-1)!!$.

To prove this result we will combine a combinatorial argument (to reduce $\mathbb{E}S_n^{2m}$ to $(2m-1)!!$) with an integration-by-parts argument (to reduce μ_{2m}^N to $(2m-1)!!$). For the combinatorial argument we note that the double factorial counts the number of ways to partition $\{1,\ldots,2m\}$ into m different pairs, that is to write

$$\{1,\ldots,2m\} = \{x_1,y_1\} \cup \cdots \cup \{x_m,y_m\}$$

where $x_1,\ldots,x_m,y_1,\ldots,y_m$ is a permutation of $1,\ldots,2m$ and $x_k < y_k$ for all k. For example, if $m = 2$ then we have the following pairings of $1,2,3,4$:

$$\{1,2\},\{3,4\} \qquad \{1,3\},\{2,4\} \qquad \{1,4\},\{2,3\}$$

and $3!! = 3$. And if $m = 3$ then we have $5!! = 15$ total pairings:

$$\begin{array}{lll}
\{1,2\},\{3,4\},\{5,6\} & \{1,2\},\{3,5\},\{4,6\} & \{1,2\},\{3,6\},\{4,5\} \\
\{1,3\},\{2,4\},\{5,6\} & \{1,3\},\{2,5\},\{4,6\} & \{1,3\},\{2,6\},\{4,5\} \\
\{1,4\},\{2,3\},\{5,6\} & \{1,4\},\{2,5\},\{3,6\} & \{1,4\},\{2,6\},\{3,5\} \\
\{1,5\},\{2,3\},\{4,6\} & \{1,5\},\{2,4\},\{3,6\} & \{1,5\},\{2,6\},\{3,4\} \\
\{1,6\},\{2,3\},\{4,5\} & \{1,6\},\{2,4\},\{3,5\} & \{1,6\},\{2,5\},\{3,4\}.
\end{array}$$

This relationship holds for all m:

Lemma 1 (pair partitions)**.** *The number of partitions of* $2m$ *into pairs equals* $(2m-1)!!$.

Proof The relationship between the double factorial and the number of pair partitions can be deduced by a very simple inductive argument. When $m = 1$ we have exactly $1!! = 1$ pairings. If we assume the number of pairings of $2,4,\ldots,2m$ are given by $3!!,5!!,\ldots,(2m-1)!!$ then we deduce the number of pairings of $2(m+1)$ as follows. We can pair the first number (1) with a choice of the $2m+1$ other numbers $(2, 3, \ldots,$ or $2m+2)$. But then by the induction hypothesis the remaining $2m$ numbers have exactly $(2m-1)!!$ pairings. Thus we have

$$\underbrace{(2m+1)}_{\text{Choices for 1}} \times \underbrace{(2m-1)!!}_{\text{Choices for pairing remaining } 2m \text{ elements}} = (2m+1)!!$$

where the last equality follows from the recurrence for double factorials.

■

We now show the moments of a Gaussian are also given in terms of the double factorial:

Lemma 2 (Gaussian moments)**.**

$$\mu_k^N := \int_{-\infty}^{\infty} x^k \frac{\mathrm{e}^{-x^2/2}}{\sqrt{2\pi}} \mathrm{d}x = \begin{cases} 0 & \textit{if } k \textit{ is odd} \\ (k-1)!! & \textit{if } k \textit{ is even} \end{cases}$$

Proof The case of k odd follows from symmetry. For even k we use a proof by induction. The base case of $k=0$ follows using the classic trick of computing the integral of a Gaussian by turning it into a 2D integral, using polar coordinates $(r^2 = x^2 + y^2)$ and the change of variables $t = r^2/2$:

$$\begin{aligned} \left(\int_{-\infty}^{\infty} \mathrm{e}^{-x^2/2}\mathrm{d}x\right)^2 &= \left(\int_{-\infty}^{\infty} \mathrm{e}^{-x^2/2}\mathrm{d}x\right)\left(\int_{-\infty}^{\infty} \mathrm{e}^{-y^2/2}\mathrm{d}y\right) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathrm{e}^{-(x^2+y^2)/2}\mathrm{d}x\mathrm{d}y \\ &= \int_0^{\infty}\int_0^{2\pi} r\mathrm{e}^{-r^2/2}\mathrm{d}r\mathrm{d}\theta = 2\pi\int_0^{\infty} \mathrm{e}^{-t}\mathrm{d}t = 2\pi, \end{aligned}$$

hence

$$\mu_0^N = \frac{\int_{-\infty}^{\infty} \mathrm{e}^{-x^2/2}\mathrm{d}x}{\sqrt{2\pi}} = 1.$$

For even $k > 0$ assume it is true for all smaller k. Then we can integrate-by-parts to deduce:

$$\begin{aligned} \sqrt{2\pi}\mu_k^N &= \int_{-\infty}^{\infty} x^k \mathrm{e}^{-x^2/2}\mathrm{d}x = \int_{-\infty}^{\infty} x^{k-1} \underbrace{x\mathrm{e}^{-x^2/2}}_{=-\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{e}^{-x^2/2}} \mathrm{d}x \\ &= [-x^{k-1}\mathrm{e}^{-x^2/2}]_{-\infty}^{\infty} + (k-1)\int_{-\infty}^{\infty} x^{k-2}\mathrm{e}^{-x^2/2}\mathrm{d}x = \sqrt{2\pi}(k-1)\mu_{k-2}^N. \end{aligned}$$

I.e., $\mu_k^N = (k-1)\mu_{k-2}^N$, but this is exactly the formula that we showed $(k-1)!!$ satisfies above (if we write $k = 2m$).

■

I.3 Proof of CLT using Method of Moments

We now want to show that the moments $\mathbb{E}S_n^k$ for

$$S_n := \frac{X_1 + \cdots + X_n}{\sqrt{n}}.$$

are also given by the double factorial as $n \to \infty$ and hence match the moments of a Normal Distribution. We show this by relating the moments to the number of pair partitions:

Theorem 1 (CLT)**.** *Let $X_1, X_2, \ldots, X_n$ be iid variables with $\mathbb{E}X_j = 0$, $\mathbb{E}X_j^2 = 1$, and all other moments are finite. Then for all k we have*

$$\mathbb{E}S_n^k \underset{n\to\infty}{\longrightarrow} \mu_k^N.$$

Proof We are going to present a slightly unusual argument that mimics the upcoming proof of random matrix universality as much as possible. The X_j are iid and we can write their moments as

$$\mu_k = \mathbb{E}X_j^k$$

where $\mu_0 = 1$ and by assumption $\mu_1 = 0$ and $\mu_2 = 1$.

We first note we can naively expand

$$(X_1 + \cdots + X_n)^k = \sum_{i_1,\ldots,i_k=1}^{n} X_{i_1} \cdots X_{i_k},$$

i.e., a sum with n^k terms. Since we know X_i commute, one angle of proof is to simplify this using the multinomial expansion to combine the terms where the different i_ℓ match, but instead we are going to break up the sum according to whether the components are the same or different, which can be identified with partitions of $\{1, \ldots, n\}$.

Let's walk through some simple cases. For $k = 1$ the result is trivial:

$$\mathbb{E}S_n = n^{-1/2} \sum_{i=1}^{n} \mathbb{E}X_i = 0.$$

For $k = 2$ we consider

$$\mathbb{E}S_n^2 = n^{-1} \sum_{i_1,i_2=1}^{n} \mathbb{E}[X_{i_1} X_{i_2}]$$

by breaking up the sum over two different cases: either $i_1 = i_2$ or $i_1 \neq i_2$. We can associate these with partitions of 2 numbers as either $\{1, 2\}$ (when they match) or $\{1\}, \{2\}$ (when they differ). Here is a table of which partitions correspond to which terms, and how many of each type of term we have, where now we assume $i_1 \neq i_2$:

Partition	Term	Count
$\{1\}, \{2\}$	$\mathbb{E}[X_{i_1} X_{i_2}] = \mathbb{E}X_{i_1}\mathbb{E}X_{i_2} = \mu_1^2 = 0$	$n(n-1) = n^2 + O(n)$
$\{1, 2\}$	$\mathbb{E}X_{i_1}^2 = \mu_2 = 1$	n

Note since $n^2 = n + n(n-1)$ we have accounted for all terms. But the only terms that are non-zero are those associated with $\{1, 2\}$. Thus we find the second moment reduces to

$$\mathbb{E}S_n^2 = n^{-1} \sum_{i_1=1}^{n} \mathbb{E}X_{i_1}^2 = 1.$$

For $k = 3$ we consider

$$\mathbb{E}S_n^3 = n^{-3/2} \sum_{i_1,i_2,i_3=1}^{n} \mathbb{E}[X_{i_1} X_{i_2} X_{i_3}]$$

by breaking up the cases according to the different partitions of 3 elements based on which terms match, where now we assume $i_1 \neq i_2 \neq i_3$:

Partition	Term	Count
$\{1\}, \{2\}, \{3\}$	$\mathbb{E}[X_{i_1} X_{i_2} X_{i_3}] = \mu_1^3 = 0$	$n(n-1)(n-2) = n^3 + O(n^2)$
$\{1, 2\}, \{3\}$	$\mathbb{E}[X_{i_1}^2 X_{i_3}] = \mu_2\mu_1 = 0$	$n(n-1)$
$\{1, 3\}, \{2\}$	$\mathbb{E}[X_{i_1}^2 X_{i_2}] = \mu_2\mu_1 = 0$	$n(n-1)$
$\{1\}, \{2, 3\}$	$\mathbb{E}[X_{i_1} X_{i_2}^2] = \mu_1\mu_2 = 0$	$n(n-1)$
$\{1, 2, 3\}$	$\mathbb{E}X_{i_1}^3 = \mu_3$	n

We have accounted for all $n^3 = n(n-1)(n-2) + 3n(n-1) + n$ terms and in this case all terms vanish. We do one last explicit example: for $n = 4$ where we assume $i_1 \neq i_2 \neq i_3 \neq i_4$ we have the following cases:

Partition	Term	Count
$\{1\},\{2\},\{3\},\{4\}$	$\mathbb{E}[X_{i_1}X_{i_2}X_{i_3}X_{i_4}] = \mu_1^4 = 0$	$n(n-1)(n-2)(n-3) = n^4 + O(n^2)$
$\{1,2\},\{3\},\{4\}$	$\mathbb{E}[X_{i_1}^2 X_{i_3}X_{i_4}] = \mu_2\mu_1^2 = 0$	$n(n-1)(n-2) = n^3 + O(n^2)$
$\{1,3\},\{2\},\{4\}$	$\mathbb{E}[X_{i_1}^2 X_{i_2}X_{i_4}] = \mu_2\mu_1^2 = 0$	$n(n-1)(n-2) = n^3 + O(n^2)$
$\{1,4\},\{2\},\{3\}$	$\mathbb{E}[X_{i_1}^2 X_{i_2}X_{i_3}] = \mu_2\mu_1^2 = 0$	$n(n-1)(n-2) = n^3 + O(n^2)$
$\{1\},\{2,3\},\{4\}$	$\mathbb{E}[X_{i_1}X_{i_2}^2 X_{i_4}] = \mu_1\mu_2\mu_1 = 0$	$n(n-1)(n-2) = n^3 + O(n^2)$
$\{1\},\{2,4\},\{3\}$	$\mathbb{E}[X_{i_1}X_{i_2}^2 X_{i_3}] = \mu_1\mu_2\mu_1 = 0$	$n(n-1)(n-2) = n^3 + O(n^2)$
$\{1\},\{2\},\{3,4\}$	$\mathbb{E}[X_{i_1}X_{i_2}X_{i_3}^2] = \mu_1^2\mu_2 = 0$	$n(n-1)(n-2) = n^3 + O(n^2)$
$\{1,2\},\{3,4\}$	$\mathbb{E}[X_{i_1}^2 X_{i_3}^2] = \mu_2^2 = 1$	$n(n-1) = n^2 + O(n)$
$\{1,3\},\{2,4\}$	$\mathbb{E}[X_{i_1}^2 X_{i_2}^2] = \mu_2^2 = 1$	$n(n-1) = n^2 + O(n)$
$\{1,4\},\{2,3\}$	$\mathbb{E}[X_{i_1}^2 X_{i_2}^2] = \mu_2^2 = 1$	$n(n-1) = n^2 + O(n)$
$\{1,2,3\},\{4\}$	$\mathbb{E}[X_{i_1}^3 X_{i_4}] = \mu_3\mu_1 = 0$	$n(n-1) = n^2 + O(n)$
$\{1,2,4\},\{3\}$	$\mathbb{E}[X_{i_1}^3 X_{i_3}] = \mu_3\mu_1 = 0$	$n(n-1) = n^2 + O(n)$
$\{1\},\{2,3,4\}$	$\mathbb{E}[X_{i_1}X_{i_2}^3] = \mu_1\mu_3 = 0$	$n(n-1) = n^2 + O(n)$
$\{1,2,3,4\}$	$\mathbb{E}X_{i_1}^4 = \mu_4$	n

The only non-zero terms are those of pair partitions (there are $n^2 + O(n)$ of those) and a partition with 4 elements (there are only n of those). Hence we deduce:

$$\begin{aligned}
\mathbb{E}S_n^4 &= n^{-2} \sum_{\text{pair partitions}} \underbrace{n(n-1)}_{\text{terms per partition}} \\
&\quad + n^{-2} \sum_{\text{paritions with 4 elements}} \underbrace{n}_{\text{terms per partition}} \mu_4 \\
&= n^{-2} \underbrace{3}_{\text{number of pair partitions}} (n^2 + O(n)) + n^{-1}\mu_4 \\
&\xrightarrow[n\to\infty]{} 3!!
\end{aligned}$$

At this point the argument is clear for general k. There are $n^k + O(n^{k-1})$ terms where each element differ, corresponding to the partition $\{1\},\dots,\{k\}$, but because of independence these all contribute zero because independence guarantees:

$$\mathbb{E}[X_{i_1}\cdots X_{i_k}] = \mu_1^k = 0$$

Each partition with a single pair has associated with it $n^{k-1} + O(n^{k-2})$ terms, but these also vanishes by the same argument since there are terms that appear once. For even k this continues until we reach exactly $k/2$ pairs, the partition that has no singletons, in which case the moments are

$$\mathbb{E}[X_{i_1}^2 \cdots X_{i_{k/2}}^2] = \mathbb{E}X_{i_1}^2 \cdots \mathbb{E}X_{i_{k/2}}^2 = \mu_2^{k/2} = 1,$$

and each possible pairing has $n(n-1)\cdots(n-k/2) = n^{k/2} + O(n^{k/2-1})$ terms. Thus the total is

$$\begin{aligned}
\mathbb{E}S_n^k &= n^{-k/2} \sum_{\text{pair partitions of } k} \underbrace{[n^{k/2} + O(n^{k/2-1})]}_{\text{number of terms per partition}} + O(n^{-1}) \\
&= (k-1)!! + O(n^{-1}) \\
&\xrightarrow[n\to\infty]{} (k-1)!!
\end{aligned}$$

When k is odd, the non-vanishing partition with the most number of terms correspond to a single 3 elements and $(k-3)/2$ pairs. As there are only $n(n-1)\cdots(n-(k-3)/2) = n^{(k-1)/2} + O(n^{(k-3)/2})$ of these, the scaling $n^{-k/2}$ dominates and we have

$$\mathbb{E}S_n^k = n^{-1/2} + O(n^{-3/2}) \underset{n\to\infty}{\longrightarrow} 0.$$

■

Remark Note the assumption of finite moments is essential. A classic counter example is if X_k have a heavy-tail distribution like the Cauchy distribution:

$$\frac{1}{\pi}\frac{1}{x^2+1}.$$

Not even the first moment exists, that is, this distribution has no mean! (Symmetry may make one think that the mean should be zero, but if we average samples it never converges.) If we take iid samples X_k of the Cauchy distribution we know that

$$\frac{X_1 + \cdots + X_n}{n}$$

is the same Cauchy distribution. (Note the division by n, not $\sqrt{n}$, which differs from the CLT above.)

I.4 Poster projects

While I expect most projects to be on random matrices, there are some interesting angles for projects that focus more on the CLT. Here are a few examples:

1. Give a detailed explanation of the Method of Moments and a comparison to an alternative analysis-based proof using the moment generating function $\mathbb{E}e^{tX}$.
2. Explore multiplicative version of the CLT involving rescaled products of positive iid random numbers. What distribution arises as the number of products becomes infinite?
3. Exploration (either experimental or mathematical) of the CLT for heavytailed distributions like the Cauchy distribution where the Method of Moments breaks down.

Chapter II

Universality

We now turn to universality for random matrices: eigenvalue statistics of symmetric random matrices are largely independent of the distribution of the entries. This can be viewed as an analogue of the CLT, but now instead of a simple sum the manipulation of the random variables involves computation of eigenvalues. In particular, we will look at both *global statistics*, i.e., what do histograms of all eigenvalues look like, and *local statistics*, i.e., what do histograms of the largest eigenvalue look like. For the theory we focus on the global statistics (as proving results for the local statistics requires much more sophisticated mathematical tools and the formula for the distribution involves Painlevé transcendents with no exact formula), and in particular, we focus on the phenomenon that symmetric matrices with independent, identically distributed (iid) entries have global statistics that look like the *semicircle*:

$$\frac{\sqrt{4-x^2}}{2\pi} \qquad \text{for } x \in [-2,2].$$

The *semicircle distribution* can in multiple ways be viewed as a random matrix analogue of the *normal distribution* $\mathcal{N}$.

In this section we will show universality again using the *Method of Moments*: we will show that the moments of the eigenvalue statistics converge to the moments of a known distribution. For a scalar random variable X the distribution is generally encoded by the standard moments:

$$\mathbb{E}X^k$$

We will see that in the matrix case the global distribution of eigenvalues is encoded by the matrix-analogue of moments: for $A_n \in \mathbb{R}^{n\times n}$ we define

$$\mathbf{E}A_n^k := \frac{1}{n}\mathbb{E}\mathrm{Tr}\, A_n^k$$

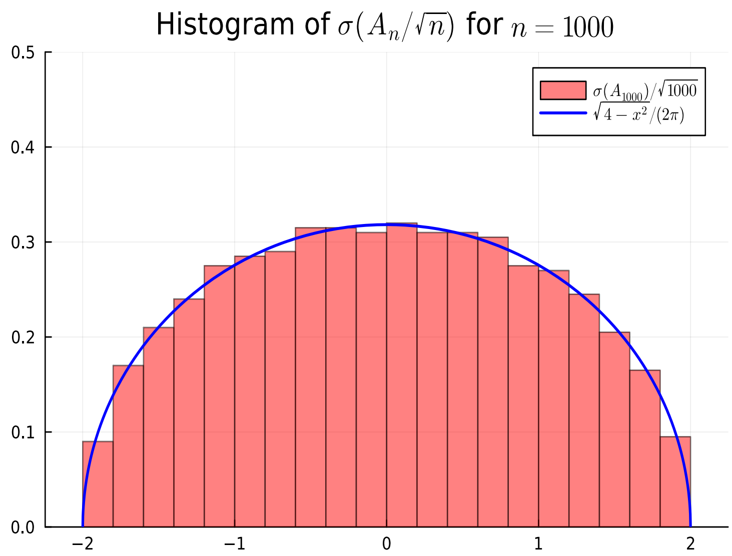
where Tr is the trace of a matrix (sum of the diagonal). It may not be obvious but these simple matrix-versions of "expectations" are equivalent to moments of the eigenvalue distribution. As the dimension of the matrix becomes large ($n \to \infty$) we will show these matrix moments tend to the moments of a semicircle.

Note that moments are only guaranteed to uniquely define a density in the special case where the probability density decays sufficiently fast, but this is something the Gaussian and semicircle both satisfy.

Further reading: Topics in Random Matrix Theory by Terry Tao gives a more advanced, higher level introduction to random matrix theory, though perhaps too high a level for most 1st year students. The field of Random Matrix Theory and universality grew out of a beautiful collaboration between theoretical physicists, statisticians, and pure mathematicians and involves the synthesis of tools between these different areas.

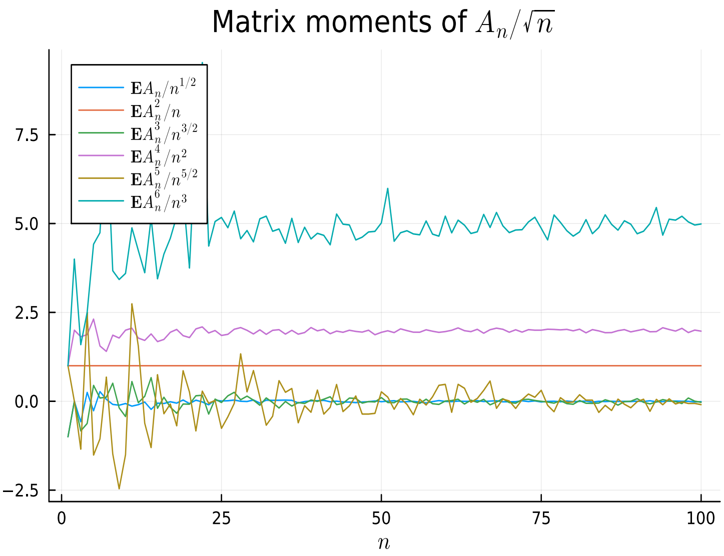
II.1 Experimental observation of universality

As a simple example, consider $A_n \in \mathbb{R}^{n\times n}$ which is symmetric ($A_n = A_n^\top$) and whose entries on and above the diagonal are iid random variables taking values ± 1 with equal probability (and entries below the diagonal defined by symmetry). Each sample of A_n will have n eigenvalues, which we denote $\sigma(A_n) = \{\lambda_1, \ldots, \lambda_n\}$ (for concreteness we can assume they are sorted from smallest to largest). If we plot a histogram of rescaled eigenvalues $\sigma(A_n/\sqrt{n}) = \left\{\frac{\lambda_1}{\sqrt{n}}, \ldots, \frac{\lambda_n}{\sqrt{n}}\right\}$ of a *single sample* A_n we get a semicircle:



Just as in the CLT, this phenomena is independent of the distribution of the entries of A_n provided the moments are finite: we always get a semicircle (shifted and scaled to match the mean and variance).

Like the CLT we will show this observation using the Method of Moments, and again we can deduce the moments experimentally by letting n increase (a single sample per n works in this case):



Again the odd moments appear to be zero but now we might conjecture that the even moments satisfy

$$\mathbf{E}A_n^2/n \to 1, \quad \mathbf{E}A_n^4/n^2 \to 2, \quad \mathbf{E}A_n^6/n^3 \to 5.$$

Again we are left with the tasks of: (1) deducing an exact formula for these moments, and (2) proving that the semicircle has the same moments.

II.2 Wigner ensembles and global statistics

To discuss universality in the setting of random matrices we focus on symmetric matrices with iid entries, i.e., the *Wigner ensembles*:

Definition 2 (Wigner)**.** A random matrix $A_n \in \mathbb{R}^{n\times n}$ is a Wigner matrix if:

1. The entries a_{kj} for $j \geq k$ are all iid with mean 0, variance 1, and all moments are finite.
2. The matrix A_n is symmetric: $a_{kj} = a_{jk}$.

We are interested in the global distribution of the eigenvalues. A Wigner ensemble $A_n \in \mathbb{R}^{n\times n}$ has n eigenvalues $\lambda_1, \ldots, \lambda_n$ which can be viewed as n random variables in their own right. The probability density associated with their histogram is described in terms of the moments:

$$\sigma_{nk} := \frac{1}{n}\mathbb{E}[\lambda_1^k + \ldots + \lambda_n^k].$$

It's a little delicate explaining why these encode the distribution of the histogram so we will just take it for granted in what follows.

We want to show that as $n \to \infty$ that these go to the moments of the semicircle. But before that we observe that we can rewrite these moments in terms of A_n itself. Recall the following:

Proposition 1 (Cyclic property of trace)**.** *For any* $A \in \mathbb{R}^{n\times m}$ *and* $B \in \mathbb{R}^{m\times n}$ *we have*

$$\mathrm{Tr}\,[AB] = \mathrm{Tr}\,[BA]$$

for the trace $\mathrm{Tr}\, A_n = a_{11} + \cdots + a_{nn}$.

Proof We show this by direct inspection:

$$\begin{aligned}
\operatorname{Tr} AB &= \sum_{k=1}^n \boldsymbol{e}_k^\top AB\boldsymbol{e}_k = \sum_{k=1}^n \begin{bmatrix} a_{k1} & \cdots & a_{km} \end{bmatrix} \begin{bmatrix} b_{1k} \\ \vdots \\ b_{m1} \end{bmatrix} = \sum_{k=1}^n \sum_{j=1}^m a_{kj} b_{jk} \\
&= \sum_{j=1}^m \sum_{k=1}^n b_{jk} a_{kj} = \sum_{j=1}^m \begin{bmatrix} b_{j1} & \cdots & b_{jn} \end{bmatrix} \begin{bmatrix} a_{j1} \\ \vdots \\ a_{jn} \end{bmatrix} = \sum_{j=1}^m \boldsymbol{e}_j^\top BA\boldsymbol{e}_j = \operatorname{Tr} BA.
\end{aligned}$$

■

A consequence is that we can relate traces of powers of symmetric matrices to sums of powers of their eigenvalues:

Proposition 2 (Traces, powers and eigenvalues)**.** *For any symmetric $A \in \mathbb{R}^{n\times n}$ with eigenvalues $\lambda_1, \ldots, \lambda_n$ we have*

$$\operatorname{Tr} A^k = \lambda_1^k + \ldots + \lambda_n^k.$$

Proof We know that a symmetric matrix is *diagonalisable*:

$$A = V \underbrace{\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}}_{\Lambda} V^{-1}.$$

(You hopefully learned that for a symmetric matrix we can choose the eigenvector matrix V to be an orthogonal matrix, i.e., where its inverse is its transpose, but we won't need that here.) Therefore

$$A^k = V\Lambda^k V^{-1}.$$

The proposition follows since the trace is equal to the sum of the eigenvalues, which follows from the cyclic property:

$$\operatorname{Tr} A^k = \operatorname{Tr} V(\Lambda^k V^{-1}) = \operatorname{Tr} (\Lambda^k V^{-1})V = \operatorname{Tr} \Lambda^k = \lambda_1^k + \ldots + \lambda_n^k.$$

■

Thus we deduce

$$\sigma_{nk} = \frac{1}{n}\mathbb{E}[\operatorname{Tr} A_n^k] = \mathbf{E}A_n^k.$$

We now have a plan of attack mirroring the CLT: (1) deduce a formula for the limit of $\sigma_{nk} = \mathbf{E}A_n^k$ as $n \to \infty$ and (2) show these match the moments of the semicircle distribution.

II.3 Moments of the semicircle distribution

We now establish the moments of the semicircle distribution, using a combinatorial construction with a similar flavour (but not the same!) as the relationship between the normal distribution and double factorials. In the case of the semicircle the key object are the Catalan numbers C_m which play a very similar role as $(2m-1)!!$:

Definition 3 (Catalan numbers)**.**

$$C_m := \frac{1}{m+1}\binom{2m}{m} = \frac{(2m)!}{(m+1)!m!}.$$

We can compute the first few:

$$C_0 = C_1 = 1, C_2 = \frac{4!}{3!2!} = 2, C_3 = \frac{6!}{4!3!} = 5,$$

and indeed these match our experimental prediction for the even moments of the global distribution of random matrix eigenvalues.

A key property we will use is that Catalan numbers satisfy a simple recurrence:

Lemma 3 (Catalan recurrence)**.**

$$C_{m+1} = \sum_{k=0}^{m} C_k C_{m-k}.$$

Proof This proof is a quite involved generating function calculation. Going through it could be the basis of a poster project.

■

Catalan numbers are equivalent to a wide range of different combinatorial objects including (1) non-crossing partitions of m, (2) the number of Dyck words of length $2m$, (3) the number of full binary trees with $m+1$ leaves, etc.

Here we are interested in the fact that they tell us exactly how many ways to partition $\{1, \ldots, 2m\}$ into pairs $\{x_j, y_j\}$ (like the Gaussian moments), but now with the restriction that the pairs are *non-crossing*: for $k \neq j$ we never have $x_k < x_j < y_k < y_j$. For example, when $m = 2$ we have $C_2 = 4!/(3!2!) = 2$ non-crossing pairings:

$$\{1,2\}, \{3,4\} \qquad \{1,4\}, \{2,3\},$$

which excludes the pairing $\{1,3\}, \{2,4\}$ since $1 < 3 < 4 < 2$. (Note in the second non-crossing pair the second pair lies completely within the first pair, so they don't cross.) When $m = 3$ we have $C_3 = 6!/(4!3!) = 5$ non-crossing pairings:

$$\begin{array}{lll} \{1,2\}, \{3,4\}, \{5,6\} & \{1,2\}, \{3,6\}, \{4,5\} & \\ \{1,4\}, \{2,3\}, \{5,6\} & \{1,6\}, \{2,3\}, \{4,5\} & \{1,6\}, \{2,5\}, \{3,4\}. \end{array}$$

We now show the general case:

Lemma 4 (non-crossing pair partitions)**.** *The number of partitions of $2m$ into non-crossing pairs equals C_m.*

Proof

We will do a proof by induction similar to that showing the number of all pair partitions is $(2m-1)!!$. The case of $C_1 = 1$ is immediate (there is only one way to pair two numbers). If we assume the number of pairings of $2, 4, \ldots, 2m$ are given by $C_1, C_2, \ldots, C_m$ then we deduce the number of pairings of $2(m+1)$ as follows:

1. If 1 is paired with 2, there are $2m$ remaining numbers to pair, i.e., C_m possibilities.
2. The number 1 cannot be paired with 3 as then there is no possibility of pairing 2 with another number without crossing with $\{1,3\}$.
3. If 1 is paired with 4 then $\{2,3\}$ must be a pair and there are $2(m-1)$ remaining numbers to pair.
4. The number 1 cannot be paired with 5 since we cannot pair $\{2,3,4\}$ without having a left over.

Thus we see the pattern: 1 cannot be paired with any odd number since then there will always be one left over. Otherwise, if 1 is paired with an even number $2k$ we have to independently pair the $2(k-1)$ numbers $\{2,\ldots,2k-1\}$ and the $2(m+1-k)$ numbers $\{2k+1,\ldots,2(m+1)\}$. Thus there are $C_{k-1}C_{m+1-k}$ possibilities.

Putting all this together we have the total number of pairings:

$$\underbrace{C_m}_{\{1,2\}\text{ is a pair}} + \underbrace{C_{m-1}}_{\{1,4\}\text{ is a pair}} + \underbrace{C_2C_{m-2}}_{\{1,6\}\text{ is a pair}} + \underbrace{C_3C_{m-3}}_{\{1,8\}\text{ is a pair}} + \cdots + \underbrace{C_m}_{\{1,2(m+1)\}\text{ is a pair}}.$$

using the fact that $C_0 = C_1 = 1$ this reduces to

$$\sum_{k=0}^{m} C_k C_{m-k} = C_{m+1}$$

by the above recurrence, and hence the result is true by induction.

■

We now establish the moments of the semicircle distribution are also given by Catalan numbers:

Lemma 5 (semicircle moments)**.**

$$\int_{-2}^{2} x^k \frac{\sqrt{4-x^2}}{2\pi} \mathrm{d}x = \begin{cases} 0 & \textit{if } k \textit{ is odd} \\ C_{k/2} & \textit{if } k \textit{ is even} \end{cases}$$

Proof The case of k odd follows from symmetry. For even k we use the change of variables $x = 2\cos\theta$ to deduce:

$$\begin{aligned}\int_{-2}^{2} x^k \frac{\sqrt{4-x^2}}{2\pi} \mathrm{d}x &= \int_0^{\pi} (2\cos\theta)^k \frac{\sqrt{4-(2\cos\theta)^2}}{2\pi} 2\sin\theta \mathrm{d}\theta \\ &= \frac{2^{k+1}}{\pi} \int_0^{\pi} \cos^k\theta \sin^2\theta \mathrm{d}\theta = \frac{2^{k+1}}{\pi} \int_0^{\pi} \cos^k\theta (1-\cos^2\theta) \mathrm{d}\theta \\ &= \frac{2^{k+1}}{\pi} \left[\int_0^{\pi} \cos^k\theta \mathrm{d}\theta - \int_0^{\pi} \cos^{k+2}\theta \mathrm{d}\theta \right].\end{aligned}$$

We claim that

$$\int_0^{\pi} \cos^{2m}\theta \mathrm{d}\theta = \frac{\pi}{2^{2m}} \binom{2m}{m}$$

which can be seen, eg., by writing

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

and using the binomial theorem. We leave the remaining details of this proof as an exercise.

■

II.4 Proof of the semicircle law

We now show for Wigner ensembles that the matrix moments

$$\mathbf{E}\left[(A_n/\sqrt{n})^k\right] = \frac{\mathbf{E}A_n^k}{n^{k/2}}$$

tend to the Catalan numbers as $n \to \infty$. We first establish a formula for the trace of powers of a (non-random) matrix which we write the entries as

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}.$$

Proposition 3 (Trace of powers)**.** *For any k we have*

$$\operatorname{Tr} A^k = \sum_{i_1,\ldots,i_k=1}^{n} a_{i_1 i_2} a_{i_2 i_3} \cdots a_{i_k i_1}.$$

Proof The case of $k = 1$ is trivial. We first deduce a formula for the j-th column of A^k. We compute the first few:

$$\begin{aligned} A\boldsymbol{e}_j &= \begin{bmatrix} a_{1j} \\ \vdots \\ a_{nj} \end{bmatrix}, \qquad A^2\boldsymbol{e}_j = \begin{bmatrix} \sum_{i_2=1}^n a_{1i_2} a_{i_2 j} \\ \vdots \\ \sum_{i_2=1}^n a_{ni_2} a_{i_2 j} \end{bmatrix}, \\ A^3\boldsymbol{e}_j &= \begin{bmatrix} \sum_{i_2,i_3} a_{1i_2} a_{i_2 i_3} a_{i_3 j} \\ \vdots \\ \sum_{i_1,i_2} a_{ni_1} a_{i_1 i_2} a_{i_2 j} \end{bmatrix}. \end{aligned}$$

By induction (exercise) we can thus deduce that the j-th column of A^k is given by

$$A^k\boldsymbol{e}_j = \begin{bmatrix} \sum_{i_2,\ldots,i_k} a_{1i_2} a_{i_2 i_3} \cdots a_{i_k j} \\ \vdots \\ \sum_{i_2,\ldots,i_k} a_{ni_2} a_{i_2 i_3} \cdots a_{i_k j} \end{bmatrix}.$$

But now we just sum over j to get the trace:

$$\sum_{j=1}^n \boldsymbol{e}_j^\top A^k \boldsymbol{e}_j = \sum_{j=1}^n \sum_{i_2,\ldots,i_k} a_{ji_2} a_{i_2 i_3} \cdots a_{i_k j} = \sum_{i_1,\ldots,i_k=1}^n a_{i_1 i_2} a_{i_2 i_3} \cdots a_{i_k i_1}.$$

■

We arrive at the main result: the rescaled matrix moments that encode the global distribution of eigenvalues tend to the Catalan numbers, i.e., the moments of a semicircle:

Theorem 2 (semicircle law)**.** *For any Wigner ensemble A_n we have*

$$\frac{\mathbf{E}A_n^k}{n^{k/2}} \xrightarrow[n\to\infty]{} \begin{cases} C_{k/2} & \textit{if } k \textit{ is even} \\ 0 & \textit{if } k \textit{ is odd} \end{cases}.$$

Proof We will only give a sketch of the argument, leaving the technical details to a possible poster project. The entries are iid and we can write their moments as

$$\mu_k := \mathbb{E}a_{ij}^k$$

where $\mu_0 = 1$ and by assumption $\mu_1 = 0$ and $\mu_2 = 1$.

Let's look at some low order cases using the explicit expansion of the trace of powers of a matrix. When $k = 1$ the result is trivial:

$$\frac{\mathbf{E}A_n}{\sqrt{n}} = \frac{1}{n^{3/2}} \sum_{i=1}^{n} \mathbb{E}a_{ii} = 0.$$

For the $k = 2$ case we first write

$$\frac{\mathbf{E}A_n^2}{n} = \frac{1}{n^2} \sum_{i_1,i_2=1}^{n} \mathbb{E}[a_{i_1i_2}a_{i_2i_1}].$$

Now this clearly simplifies, but to prepare for the general case we again break the sum into two different cases: either the two terms equal ($a_{i_1i_2} = a_{i_1i_2}$ or $a_{i_1i_2} = a_{i_2i_1}$) or they differ. We associate these two cases with the partitions $\{1,2\}$ (terms match) and $\{1\},\{2\}$ (terms differ). But in this case all terms fall into the matching category:

Partition	Term	Count
$\{1\},\{2\}$	Not possible!	0
$\{1,2\}$	$\mathbb{E}[a_{i_1i_2}a_{i_2i_1}] = \mathbb{E}a_{i_1i_2}^2 = \mu_2 = 1$	n^2

We thus have

$$\frac{\mathbf{E}A_n^2}{n} = \frac{1}{n^2} \sum_{i_1,i_2=1}^{n} \mathbb{E}[\underbrace{a_{i_1i_2}a_{i_2i_1}}_{\text{always match}}] = 1 = C_1.$$

For $k = 3$ we consider different cases of writing

$$\frac{\mathbf{E}A_n^3}{n^{3/2}} = \frac{1}{n^{5/2}} \sum_{i_1,i_2,i_3=1}^{n} \mathbb{E}[a_{i_1i_2}a_{i_2i_3}a_{i_3i_1}].$$

We want to think of the ways in which each of these three terms can be equal (note we are interested in the terms matching, *not* the indices). This has the same number of ways as partitioning a set of 3. Assuming now $i_1 \neq i_2 \neq i_3$ we note $a_{i_1i_2} \neq a_{i_2i_3} \neq a_{i_3i_1}$ (of course since they are random variables they might accidentally be equal, so here I just mean that they aren't guaranteed to be the same by symmetry). Thus we have the following types of terms, each associated with the specified partition of 3:

Partition	Term	Count
$\{1\},\{2\},\{3\}$	$\mathbb{E}[a_{i_1i_2}a_{i_2i_3}a_{i_3i_1}] = \mu_1^3 = 0$	$n(n-1)(n-2) = n^3 + O(n^2)$
$\{1,2\},\{3\}$	$\mathbb{E}[a_{i_1i_2}a_{i_2i_1}a_{i_1i_1}] = \mathbb{E}[a_{i_1i_2}^2 a_{i_1i_1}] = \mu_2\mu_1 = 0$	$n(n-1) = n^3 + O(n^2)$
$\{1,3\},\{2\}$	$\mathbb{E}[a_{i_1i_2}a_{i_2i_2}a_{i_2i_1}] = \mathbb{E}[a_{i_1i_2}^2 a_{i_2i_2}] = \mu_2\mu_1 = 0$	$n(n-1) = n^3 + O(n^2)$
$\{1\},\{2,3\}$	$\mathbb{E}[a_{i_1i_1}a_{i_1i_3}a_{i_3i_1}] = \mathbb{E}[a_{i_1i_1}a_{i_1i_3}^2] = \mu_1\mu_2 = 0$	$n(n-1) = n^3 + O(n^2)$
$\{1,2,3\}$	$\mathbb{E}[a_{i_1i_1}a_{i_1i_1}a_{i_1i_1}] = \mathbb{E}a_{i_1i_1}^3 = \mu_3$	n

Note that they combine to give all n^3 possibilities. But the only non-zero moments are those associated with $\{1,2,3\}$, i.e., when all three terms match. Thus we can collapse the sum to be:

$$\frac{\mathbf{E}A_n^3}{n^{3/2}} = \frac{1}{n^{5/2}} \sum_{i_1=1}^{n} \mathbb{E}[a_{i_1i_1}a_{i_1i_1}a_{i_1i_1}] = \frac{\mu_3}{n^{3/2}} \xrightarrow[n\to\infty]{} 0.$$

Let's do one more example. For $k=4$ we have:

$$\frac{\mathbf{E}A_n^4}{n^2} = \frac{1}{n^3} \sum_{i_1,i_2,i_3,i_4=1}^{n} \mathbb{E}[a_{i_1i_2}a_{i_2i_3}a_{i_3i_4}a_{i_4i_1}]$$

We can again divide the sum up by which terms are equal, which are one-to-one with the number of partitions of 4 (here the restrictions on when i_α can be equal is quite complicated so we leave the count as an exercise, the first one being the most challenging!):

Partition	Term	Count
$\{1\},\{2\},\{3\},\{4\}$	$\mathbb{E}[a_{i_1i_2}a_{i_2i_3}a_{i_3i_4}a_{i_4i_1}] = 0$	$(n+1)n(n-1)(n-2) = n^4 + O(n^3)$
$\{1,2\},\{3\},\{4\}$	Not possible!	0
$\{1,3\},\{2\},\{4\}$	$\mathbb{E}[a_{i_1i_2}a_{i_2i_2}a_{i_2i_1}a_{i_1i_1}] = 0$	$n(n-1) = n^2 + O(n)$
$\{1,4\},\{2\},\{3\}$	Not possible!	0
$\{1\},\{2,3\},\{4\}$	Not possible!	0
$\{1\},\{2,4\},\{3\}$	$\mathbb{E}[a_{i_1i_1}a_{i_1i_3}a_{i_3i_3}a_{i_3i_1}] = 0$	$n(n-1) = n^2 + O(n)$
$\{1\},\{2\},\{3,4\}$	Not possible!	0
$\{1,2\},\{3,4\}$	$\mathbb{E}[a_{i_1i_2}a_{i_2i_1}a_{i_1i_3}a_{i_3i_1}] = \mu_2^2 = 1$	$n^2(n-1) = n^3 + O(n^2)$
$\{1,3\},\{2,4\}$	Not possible!	0
$\{1,4\},\{2,3\}$	$\mathbb{E}[a_{i_1i_2}a_{i_2i_3}a_{i_3i_2}a_{i_2i_1}] = \mu_2^2 = 1$	$n^2(n-1) = n^3 + O(n^2)$
$\{1,2,3\},\{4\}$	Not possible!	0
$\{1,2,4\},\{3\}$	Not possible!	0
$\{1,3,4\},\{2\}$	Not possible!	0
$\{1\},\{2,3,4\}$	Not possible!	0
$\{1,2,3,4\}$	$\mathbb{E}[a_{i_1i_2}a_{i_2i_1}a_{i_1i_2}a_{i_2i_1}] = \mu_4$	n^2

One could go through case-by-case and argue why the "Not possible!" cases aren't possible. For example, we know $\{1,2\},\{3\},\{4\}$ cannot happen as if the first two terms are equal they must be of the form $a_{i_1i_2}a_{i_2i_1}$, but that implies the last two terms also have to be equal. But we don't even need to do this: when you add up all the possible terms we get exactly n^4, and hence there cannot be any more.

Thus the moment reduces to (where the summation index indicates which type of partition the terms correspond to)

$$\frac{\mathbf{E}A_n^4}{n^2} = \frac{1}{n^3}\left[\underbrace{\sum_{\{1,2,3,4\}\text{ terms}} \mu_4}_{=n^2\mu_4} + \underbrace{\sum_{\{1,2\},\{3,4\}\text{ terms}} \mu_2^2}_{=n^3+O(n^2)} + \underbrace{\sum_{\{1,4\},\{2,3\}\text{ terms}} \mu_2^2}_{=n^3+O(n^2)}\right] \xrightarrow[n\to\infty]{} 2,$$

which matches the Catalan number C_2.

Now for the key insight that mirrors the proof the CLT: when we break apart the sum into different categories of partitions, the the largest terms that do not vanish are those corresponding to non-crossing pairs of 4, in this case

$$\{1,2\},\{3,4\} \qquad \{1,4\},\{2,3\}$$

Each of these contributes exactly $n^3 + O(n^2)$, and hence as $n \to \infty$ the moment tends to the number of non-crossing pairs of 4.

This pattern holds true for all k. When k is odd, we have n^k total terms which we can divide up according to the partitions of k numbers. But the only ones that actually contribute are those with no single element in a partition. We omit the details but what remains grows slower than $n^{k/2+1}$ and hence the moment tends to zero. When k is even, the only terms that grow precisely like $n^{k/2+1}$ are those corresponding to non-crossing pair partitions of k. Since each of these corresponds to a pair we know its expectation is

$$\mu_2^{k/2} = 1.$$

The number of terms (and therefore the sum) exactly equals the Catalan number $C_{k/2}$!

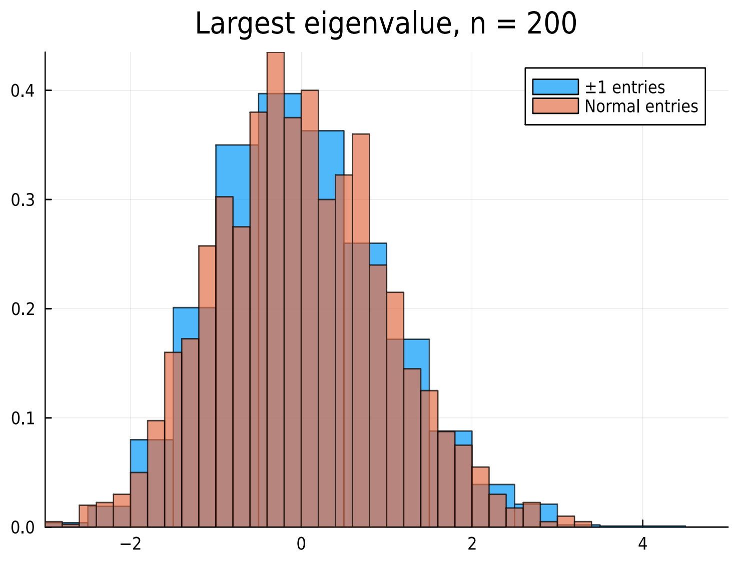
■

II.5 Local eigenvalue statistics

One can also ask about the statistics of the largest eigenvalue. Here we plot a histogram of the largest eigenvalue divided of two different Wigner ensembles (one with entries equal to ± 1 and the other with normally distributed eigenvalues). If $\lambda_k^{\max}$ denotes independent samples of the largest eigenvalue, to normalise the distribution we actually plot a histogram of

$$\frac{\lambda_k^{\max} - \mathbb{E}\lambda_k^{\max}}{\operatorname{std}(\lambda_k^{\max})}$$

where std denotes the standard deviation. We see that the two different ensembles have the same largest eigenvalue distribution:



This distribution is known as the Tracy–Widom distribution. It has no simple formula but can be written either in terms of a Painlevé function (which solves a nonlinear ODE).

II.6 Poster projects

The possibilities for projects related to universality are endless and I encourage you to think about an original idea. Here are some example projects to help guide you:

1. Give a detailed explanation of Catalan numbers and the full proof of the semicircle law.
2. Other local statistics also exhibit universality, for example, the spacing between two eigenvalues in the *bulk* (i.e. away from the edges of the semicircle). Explore this phenomena.
3. Hermitian random matrices also satisfy universality properties. How do histograms of their eigenvalue statistics (both global and local) compare to the symmetric case?
4. A *Wishart ensemble* is a random symmetric matrix $XX^\top$ where $X \in \mathbb{R}^{m\times n}$ is a rectangular matrix with random entries. It also exhibits universality but depending on the ratio $\alpha = m/n$, where the eigenvalue distributions tend to the Marchenko–Pastur distribution. Explore this phenomena and how α dictates which Marchenko–Pastur distribution arises. Does the largest eigenvalue still follow the Tracy–Widom distribution?
5. The spacing of zeros of the Riemann Zeta function on the critical line match universality, in particular, the spacing of eigenvalues in the bulk. Explore computation of zeros of Riemann Zeta functions (using eg. `scipy.special.zeta`) and compare with suitably-rescaled random matrix statistics.
6. What happens when the moments are no longer defined? I.e. what are eigenvalue statistics for matrices with heavytailed entries?
7. One can consider matrices associated with random networks, a la as discussed in Introduction to Applied Mathematics. If a network is completely connected it will be a Wigner ensemble. Explore experimentally how connected a random network needs to be to exhibit universality.

Chapter III

Free Probability

We now switch our attention and ask: given two (symmetric) random matrices $A_n, B_n \in \mathbb{R}^{n\times n}$, what are the distribution of the eigenvalues of $A_n + B_n$? If both A_n and B_n are Wigner matrices then so is their sum, so we are really interested in the case where one or both are not Wigner matrices (i.e., the distribution of the entries is not all the same or there is covariance between the entries). The rather remarkable fact is that, as $n \to \infty$, if we know the global distribution of A_n and B_n, in the form of knowing the moments $\mathbf{E}A_n^k$ and $\mathbf{E}B_n^k$, under very broad conditions we can deduce the global distribution of $A_n + B_n$, in the form of knowing the moments $\mathbf{E}(A_n + B_n)^k$, where again we use

$$\mathbf{E}f(A_n) := \frac{1}{n}\mathbb{E}\mathrm{Tr}\, f(A_n).$$

In order to make sense of this, we think of the large n limit of A_n and B_n, a limit which does not exist in a traditional sense but we can formally write as A and B, as an abstract algebra which behave simple rules. For example, since A_n and B_n do not commute, neither do A and B ($AB \neq BA$). More specifically, A and B are *freely independent* non-commutative random variables, which is defined in terms of an expectation which we denote $\mathbf{E}$, which satisfies

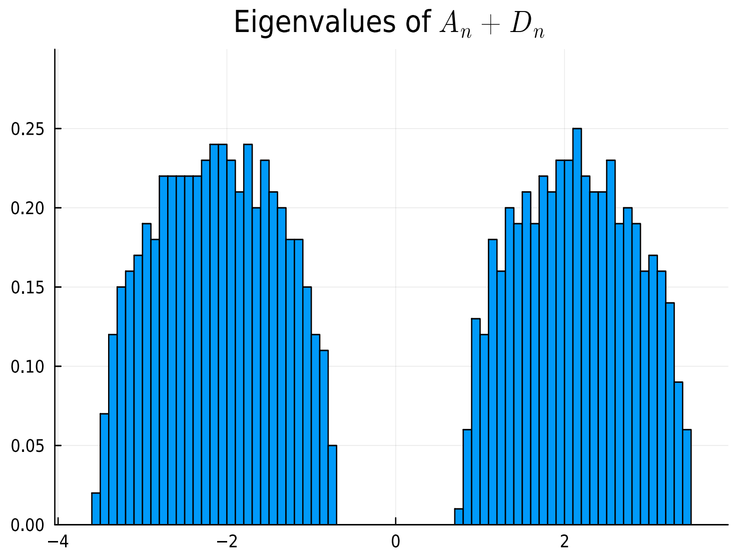
$$\mathbf{E}f(A) = \lim_{n\to\infty} \mathbf{E}f(A_n).$$

(We have seen in the section on universality that even though A_n itself doesn't really have a limit, $\mathbf{E}f(A_n)$ will converge, and in particular behave like a semicircle.) By writing down the rules of $\mathbf{E}$ we can deduce explicit formulae for the moments $\mathbf{E}$ (A+B)^k, which will tell us the large n behaviour of the eigenvalues of $A + B$. This can be thought of as a *free convolution*, a non-commutative analogue of convolution.

Further Reading Free Probability Theory by Roland Speicher is an advanced introduction to the topic whilst Random Matrix Theory by Edelman and Rao gives a gentle but brief introduction. Free Probability was introduced by Dan Voiculescu, a prominent Berkeley operator theorist, in the 1980s. The connection with random matrix theory actually came much later in the early 1990s, as well as a detailed combinatorial description of Free Probability in terms of non-crossing partitions by Speicher. It is a very modern and technical subject, and there is unfortunately limited material written for 1st year students beyond these notes. Rather exceptionally for an idea arising in pure mathematics, it has had an impact in a wide range of applications including wireless communication, quantum information, and most recently machine learning (where it has been used to study the transition from random-like behaviour of untrained networks to non-random behaviour of trained networks).

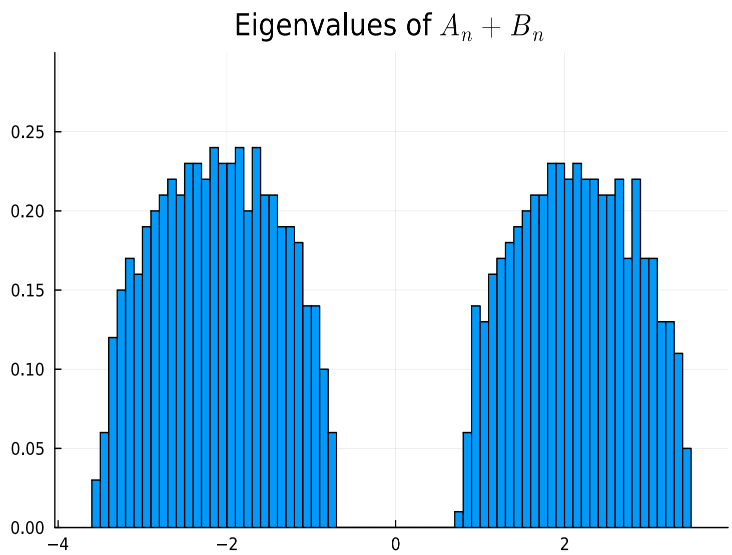
III.1 Experimental observation of free convolutions

We begin with an experimental observation of free convolution. Consider again a symmetric matrix $A_n \in \mathbb{R}^{n\times n}$ whose entries on and above the diagonal are iid random variables taking values ± 1 with equal probability. We saw in the previous chapter the global statistics are given by the semicircle. Now consider adding a diagonal matrix $D_n \in \mathbb{R}^{n\times n}$ with ± 2 on the diagonal with equal probability: what are the global statistics of $A_n + D_n$? We can explore this question numerically:



We see that we get two semicircle-like distributions, surrounding ± 2.

What's remarkable is this distribution only depends on the eigenvalues of A_n and D_n. For example, consider the case where we same randomly $Q_n \in O(n)$, i.e., a random orthogonal matrix and define $B_n := Q_n D_n Q_n^\top$. This has exactly the same eigenvalues as D_n itself. And when we add $A_n + B_n$ we get the exact same global statistics:



This is a numerical demonstration of the fact that we can, under broad conditions, deduce the global eigenvalue statistics of $A_n + B_n$ from those of A_n and B_n, and largely ignore the eigenvectors. To prove this we will again use the Method of Moments.

III.2 Free random variables

Since we aim to describe the behaviour of random matrices when the dimension becomes infinite, it's helpful to put aside viewing A and B as matrices and instead think of them more abstractly as *free random variables*, which are non-commutative random variables. With this convention, we will denote the identity element (the limit of the identity matrix I_n as $n \to \infty$) as 1. Our free random variables expectation $\mathbf{E}$ is no longer given explicitly in terms of a trace of a matrix, and instead we describe it more abstractly in terms of the properties it satisfies.

With classical probability we have the feature that the expectations of the product of independent random variable equals the product of the expectation. More generally if f and g are polynomials and X and Y are independent we know:

$$\mathbb{E}[f(X)g(Y)] = \mathbb{E}[f(X)]\mathbb{E}[g(Y)].$$

We will see this particular property *does* hold true for free probability. Where free probability differs to classical random variables is in alternating products: since X and Y are *commutative*, if we have polynomials $f_1, \ldots, f_n$ and $g_1, \ldots, g_n$ then we clearly have

$$\mathbb{E}[f_1(X)g_1(Y) \cdots f_n(X)g_n(Y)] = \mathbb{E}[f_1(X) \cdots f_n(X)]\mathbb{E}[g_1(Y) \cdots g_n(Y)],$$

but this is no longer the case when X and Y do not commute.

Free probability comes from defining how these mixed moments behave in the case where the expectation vanishes:

Definition 4 (free random variables)**.** A and B are *free random variables* equipped with an expectation $\mathbf{E}$ if for all polynomials $f_1, \ldots, f_n$ and $g_1, \ldots, g_n$ such that $\mathbf{E}f_k(A) = \mathbf{E}g_k(B) = 0$ we have

$$\mathbf{E}[f_1(A)g_1(B) \ldots f_n(A)g_n(B)] = 0,$$

where the identity element satisfies $\mathbf{E}1 = 1$.

We claim (without proof) that as $n \to \infty$ that $\frac{1}{n}\mathbb{E}\mathrm{Tr}$ behaves exactly like this for many families of random matrices.

Note that we assume linearity: for constants c, d we have

$$\mathbf{E}[cA + dB] = c\mathbf{E}A + d\mathbf{E}B.$$

It follows that (treating $\mathbf{E}A^k$ as a constant)

$$\mathbf{E}[A^k - \mathbf{E}A^k] = \mathbf{E}A^k - \mathbf{E}A^k\mathbf{E}1 = 0.$$

This simple observation guarantees that if we know the moments $\mathbf{E}A^k$ and $\mathbf{E}B^k$ we can deduce the moments $\mathbf{E}[A^kB^j]$. To see this, we first consider some of the low order cases. By subtracting off the free expectation of moments of A and B and treating $\mathbf{E}A^k, \mathbf{E}B^j$ as constants we deduce:

$$\begin{aligned} 0 &= \mathbf{E}[(A^k - \mathbf{E}A^k)(B^j - \mathbf{E}B^j)] = \mathbf{E}[A^kB^j] - \mathbf{E}[A^k\mathbf{E}B^j] - \mathbf{E}[\mathbf{E}[A^k]B^j] + \mathbf{E}[\mathbf{E}A^k\mathbf{E}B^j] \\ &= \mathbf{E}[A^kB^j] - \mathbf{E}A^k\mathbf{E}B^j \qquad \Rightarrow \qquad \mathbf{E}[A^kB^j] = \mathbf{E}A^k\mathbf{E}B^j \end{aligned}$$

which matches the classical random variables formula $\mathbb{E}[A^k B^j] = \mathbb{E}A^k \mathbb{E}B^j$. This apparent consistency with classic random variables works for one mixed moment:

$$\begin{aligned}
0 &= \mathbf{E}[(A - \mathbf{E}A)(B - \mathbf{E}B)(A - \mathbf{E}A)] \\
&= \mathbf{E}[ABA] - \mathbf{E}[AB\mathbf{E}[A]] - \mathbf{E}[A\mathbf{E}[B]A] + \mathbf{E}[A\mathbf{E}[B]\mathbf{E}[A]] \\
&\quad - \mathbf{E}[\mathbf{E}[A]BA] + \mathbf{E}[\mathbf{E}[A]B\mathbf{E}[A]] + \mathbf{E}[\mathbf{E}[A]\mathbf{E}[B]A] - \mathbf{E}[\mathbf{E}[A]\mathbf{E}[B]\mathbf{E}[A]] \\
&= \mathbf{E}[ABA] - \mathbf{E}A \underbrace{\mathbf{E}[AB]}_{=\mathbf{E}A\mathbf{E}B} - \mathbf{E}B\mathbf{E}A^2 - \mathbf{E}A \underbrace{\mathbf{E}[BA]}_{=\mathbf{E}A\mathbf{E}B} + 2\mathbf{E}[A]^2\mathbf{E}B \\
&= \mathbf{E}[ABA] - \mathbf{E}B\mathbf{E}A^2 \qquad \Rightarrow \\
\mathbf{E}[ABA] &= \mathbf{E}A^2\mathbf{E}B.
\end{aligned}$$

But that's where the equivalence ends. In particular, consider:

$$\begin{aligned}
0 &= \mathbf{E}[(A - \mathbf{E}A)(B - \mathbf{E}B)(A - \mathbf{E}A)(B - \mathbf{E}B)] \\
&= \mathbf{E}[ABAB] - \underbrace{\mathbf{E}[ABA]}_{=\mathbf{E}A^2\mathbf{E}B}\mathbf{E}B - \mathbf{E}A\underbrace{\mathbf{E}[AB^2]}_{=\mathbf{E}A\mathbf{E}B^2} + \underbrace{\mathbf{E}[AB]}_{=\mathbf{E}A\mathbf{E}B}\mathbf{E}A\mathbf{E}B \\
&\quad - \underbrace{\mathbf{E}[A^2B]}_{=\mathbf{E}A^2\mathbf{E}B}\mathbf{E}B + \mathbf{E}A^2\mathbf{E}[B]^2 + \underbrace{\mathbf{E}[AB]}_{=\mathbf{E}A\mathbf{E}[B]}\mathbf{E}A\mathbf{E}B - \mathbf{E}[A]^2\mathbf{E}[B]^2 \\
&\quad - \mathbf{E}[A]\underbrace{\mathbf{E}[BAB]}_{=\mathbf{E}A\mathbf{E}B^2} + \mathbf{E}A\mathbf{E}B\underbrace{\mathbf{E}[BA]}_{=\mathbf{E}A\mathbf{E}B} + \mathbf{E}[A]^2\mathbf{E}B^2 - \mathbf{E}[A]^2\mathbf{E}[B]^2 \\
&\quad + \mathbf{E}A\mathbf{E}B\underbrace{\mathbf{E}[AB]}_{=\mathbf{E}A\mathbf{E}B} - \mathbf{E}[A]^2\mathbf{E}[B]^2 - \mathbf{E}[A]^2\mathbf{E}[B]^2 + \mathbf{E}[A]^2\mathbf{E}[B]^2 \\
&= \mathbf{E}[ABAB] - \mathbf{E}A^2\mathbf{E}[B]^2 - \mathbf{E}[A]^2\mathbf{E}B^2 + \mathbf{E}[A]^2\mathbf{E}[B]^2 \qquad \Rightarrow \\
\mathbf{E}[ABAB] &= \mathbf{E}A^2\mathbf{E}[B]^2 + \mathbf{E}[A]^2\mathbf{E}B^2 - \mathbf{E}[A]^2\mathbf{E}[B]^2
\end{aligned}$$

which differs from the classical (commutative) case where $\mathbb{E}[ABAB] = \mathbb{E}A^2\mathbb{E}B^2$. Nevertheless, we can see the central theme emerging: $\mathbf{E}[ABAB]$ is still given in terms of a function of the input moments $\mathbf{E}A^k$ and $\mathbf{E}B^j$:

Lemma 6 (mixed moments)**.** *For $k_1, \ldots, k_n$ and $j_1, \ldots, j_n$ the mixed moments*

$$\mathbf{E}[A^{k_1}B^{j_1}\cdots A^{k_n}B^{j_n}]$$

can be expressed as functions of the moments

$$\mathbf{E}A, \mathbf{E}A^2 \ldots, \mathbf{E}A^{k_1+\cdots+k_n}, \mathbf{E}B, \mathbf{E}B^2 \ldots, \mathbf{E}B^{j_1+\cdots+j_n}.$$

Proof The proceeding example can be turned into a proof by induction. We leave this as a possible poster project. ■

Unfortunately, there is no simple formula for the relationship between the moments of A and B and those of the mixed moments, but they can be derived systematically like we did for the first four moments. Note the cyclic property of trace we can be used to simplify some calculations because we have

$$\mathbf{E}[A^{k_1}B^{j_1}\cdots A^{k_n}B^{j_n}] = \mathbf{E}[B^{j_1}\cdots A^{k_n}B^{j_n}A^{k_1}] = \mathbf{E}[B^{j_n}A^{k_1}B^{j_1}\cdots A^{k_n}].$$

For example, we can deduce all moments up to degree 4 using the above calculations:

$$\begin{aligned}
\mathbf{E}[AB] &= \mathbf{E}[BA] = \mathbf{E}A\mathbf{E}B \\
\mathbf{E}[ABA] &= \mathbf{E}[A^2B] = \mathbf{E}[BA^2] = \mathbf{E}A^2\mathbf{E}B \\
\mathbf{E}[BAB] &= \mathbf{E}[B^2A] = \mathbf{E}[AB^2] = \mathbf{E}A\mathbf{E}B^2 \\
\mathbf{E}[A^2BA] &= \mathbf{E}[ABA^2] = \mathbf{E}[A^3B] = \mathbf{E}[BA^3] = \mathbf{E}A^3\mathbf{E}B \\
\mathbf{E}[B^2AB] &= \mathbf{E}[BAB^2] = \mathbf{E}[B^3A] = \mathbf{E}[AB^3] = \mathbf{E}A\mathbf{E}B^3 \\
\mathbf{E}[AB^2A] &= \mathbf{E}[BA^2B] = \mathbf{E}[A^2B^2] = \mathbf{E}[B^2A^2] = \mathbf{E}A^2\mathbf{E}B^2 \\
\mathbf{E}[BABA] &= \mathbf{E}[ABAB] = \mathbf{E}A^2\mathbf{E}[B]^2 + \mathbf{E}[A]^2\mathbf{E}B^2 - \mathbf{E}[A]^2\mathbf{E}[B]^2.
\end{aligned}$$

Note that there 2^k moments of degree k since each "slot" can be either A or B, and we can confirm we have all of them, when we include in the two trivial cases of $\mathbf{E}A^k$ and $\mathbf{E}B^k$.

III.3 Free convolution

An implication of the previous lemma is that the expectation of all polynomials of A and B can be expressed in terms of the moments of A and B. Note the moments of $A+B$ are polynomials in mixed moments, for example, here we walk through the first few examples using the explicit formulæ for mixed moments (noting powers $(A+B)^k$ pick up every possible one of the 2^k mixed moments exactly once):

$$\begin{aligned}
\mathbf{E}[A+B] &= \mathbf{E}A + \mathbf{E}B \\
\mathbf{E}[(A+B)^2] &= \mathbf{E}A^2 + \underbrace{\mathbf{E}[AB]}_{=\mathbf{E}A\mathbf{E}B} + \underbrace{\mathbf{E}[BA]}_{=\mathbf{E}A\mathbf{E}B} + \mathbf{E}B^2 = \mathbf{E}A^2 + 2\mathbf{E}A\mathbf{E}B + \mathbf{E}B^2 \\
\mathbf{E}[(A+B)^3] &= \mathbf{E}A^3 + \underbrace{\mathbf{E}[A^2B]}_{=\mathbf{E}A^2\mathbf{E}B} + \underbrace{\mathbf{E}[ABA]}_{=\mathbf{E}A^2\mathbf{E}B} + \underbrace{\mathbf{E}[AB^2]}_{=\mathbf{E}A\mathbf{E}B^2} + \underbrace{\mathbf{E}[BA^2]}_{=\mathbf{E}A^2\mathbf{E}B} + \underbrace{\mathbf{E}[BAB]}_{=\mathbf{E}A\mathbf{E}B^2} + \underbrace{\mathbf{E}[B^2A]}_{=\mathbf{E}A\mathbf{E}B^2} + \mathbf{E}B^3 \\
&= \mathbf{E}A^3 + 3\mathbf{E}A^2\mathbf{E}B + 3\mathbf{E}AB^2 + \mathbf{E}B^3 \\
\mathbf{E}[(A+B)^4] &= \mathbf{E}A^4 + \underbrace{\mathbf{E}[A^3B]}_{=\mathbf{E}A^3\mathbf{E}B} + \underbrace{\mathbf{E}[A^2BA]}_{=\mathbf{E}A^3\mathbf{E}B} + \underbrace{\mathbf{E}[A^2B^2]}_{=\mathbf{E}A^2\mathbf{E}B^2} + \underbrace{\mathbf{E}[ABA^2]}_{=\mathbf{E}A^3\mathbf{E}B} \\
&\quad + \underbrace{\mathbf{E}[ABAB]}_{=\mathbf{E}A^2\mathbf{E}[B]^2+\mathbf{E}[A]^2\mathbf{E}B^2-\mathbf{E}[A]^2\mathbf{E}[B]^2} + \underbrace{\mathbf{E}[AB^2A]}_{=\mathbf{E}A^2\mathbf{E}B^2} \\
&\quad + \underbrace{\mathbf{E}[AB^3]}_{=\mathbf{E}A\mathbf{E}B^3} + \underbrace{\mathbf{E}[BA^3]}_{=\mathbf{E}A^3\mathbf{E}B} + \underbrace{\mathbf{E}[BA^2B]}_{=\mathbf{E}A^2\mathbf{E}B^2} + \underbrace{\mathbf{E}[BABA]}_{=\mathbf{E}A^2\mathbf{E}[B]^2+\mathbf{E}[A]^2\mathbf{E}B^2-\mathbf{E}[A]^2\mathbf{E}[B]^2} \\
&\quad + \underbrace{\mathbf{E}[BAB^2]}_{=\mathbf{E}A\mathbf{E}B^3} + \underbrace{\mathbf{E}[B^2A^2]}_{=\mathbf{E}A^2\mathbf{E}B^2} + \underbrace{\mathbf{E}[B^2AB]}_{=\mathbf{E}A\mathbf{E}B^3} + \underbrace{\mathbf{E}[B^3A]}_{=\mathbf{E}A\mathbf{E}B^3} + \mathbf{E}B^4 \\
&= \mathbf{E}A^4 + 4\mathbf{E}A^3\mathbf{E}B + 4\mathbf{E}A^2\mathbf{E}B^2 + 4\mathbf{E}A\mathbf{E}B^3 + \mathbf{E}B^4 \\
&\quad + 2(\mathbf{E}A^2\mathbf{E}[B]^2 + \mathbf{E}[A]^2\mathbf{E}B^2 - \mathbf{E}[A]^2\mathbf{E}[B]^2).
\end{aligned}$$

Note that up until the 4th moment the calculation is equivalent to classical convolution, and in particular the calcluation is an affine function of each of the "input" moments. The 4th moment is where free convolution starts to differ, and the last term is a truely nonlinear function of the moments.

Unfortunately there is no simple combinatorial formula for these expressions. But there is a miraculous equivalent way of constructing them. Consider the Stieltjes transform associated with the moments of A, i.e., the moment generating function of the form:

$$G_A(z) := \frac{1}{z} + \frac{\mathbf{E}A}{z^2} + \frac{\mathbf{E}A^2}{z^3} + \cdots .$$

I like to think of this as a "Taylor series around ∞"; actually, in a neighbourhood of ∞ in the complex plane (hence the usage of z) but this is beyond what you have learned as 1st year students. Thus it is conceptually easier to recast this as a Taylor series around the origin via the transformation:

$$\tilde{G}_A(z) := G_A(z^{-1}) = z + \sum_{k=2}^{\infty} \mathbf{E}A^{k-1} z^k.$$

Because its derivative at zero is nonzero (its 1) we know from the *Inverse Function Theorem* that $\tilde{G}_A$ is invertible for small z, thus there exists a function $\tilde{G}_A^{-1}$ so that $\tilde{G}_A(\tilde{G}_A^{-1}(w)) = w$ which also has some Taylor series:

$$\tilde{G}_A^{-1}(w) = w + \sum_{k=2}^{\infty} \tilde{r}_{A,k} w^k$$

where $\tilde{r}_{A,k}$ are some functions of $\mathbf{E}A^k$. (One can actually deduce them systematically by matching terms!) We can thence deduce that G_A itself has an inverse for small w via:

$$G_A^{-1}(w) := 1/\tilde{G}_A^{-1}(w) = \frac{1}{w} + \frac{r_{A,2}}{w^2} + \frac{r_{A,3}}{w^3} + \cdots ,$$

where $r_{A,k}$ can be deduced from $\tilde{r}_{A,k}$, since we have

$$G_A(G_A^{-1}(w)) = \tilde{G}_A(1/G_A^{-1}(w)) = \tilde{G}_A(\tilde{G}_A^{-1}(w)) = w.$$

We now define the *R-transform*:

$$R_A(w) := G_A^{-1}(w) - \frac{1}{w} = \frac{r_{A,2}}{w^2} + \frac{r_{A,3}}{w^3} + \cdots .$$

The miraculous fact is that the R-transform of $A + B$ is given by the sum of the R-transforms of A and B:

$$R_{A+B}(z) = R_A(z) + R_B(z).$$

Proving this is a very serious combinatorial problem connected to noncrossing partitions, which could be an excellent poster project.

Example (adding two semicircles) We conclude with an example where we can do everything explicitly. Suppose A and B are the large n limits of two independent Wigner ensembles. We know that $A + B$ is also the large n limit of a Wigner ensemble (but with double the variance) so will also have a semicircle eigenvalue distribution. How do we see this fact come out of free convolution? In this case the moments are identical:

$$\mathbf{E}A^k = \mathbf{E}B^k = \begin{cases} C_{k/2} & k \text{ even} \\ 0 & k \text{ odd} \end{cases} .$$

We therefore have

$$\tilde{G}_A(z) \equiv \tilde{G}_B(z) = z + \sum_{k=2}^{\infty} \mathbf{E}A^{k-1} z^k = C_0 z + C_1 z^3 + C_2 z^5 + \cdots .$$

Now comes the key trick: we can use the recurrence for Catalan numbers to deduce

$$\begin{aligned}\tilde{G}_A(z)^2 &= z^2\left[C_0^2 + (C_0C_1 + C_1C_0)z^2 + (C_0C_2 + C_1C_1 + C_1C_2)z^4 + \cdots\right] \\ &= z^2 \sum_{k=0}^{\infty} z^{2k} \underbrace{\sum_{j=0}^{k} C_j C_{k-j}}_{=C_{k+1}} = C_1 z^2 + C_2 z^4 + C_3 z^6 \cdots = \frac{\tilde{G}_A(z)}{z} - 1.\end{aligned}$$

We can now use the quadratic formula to deduce

$$\tilde{G}_A(z) = \frac{z^{-1} \pm \sqrt{z^{-2} - 4}}{2} = \frac{1 - \sqrt{1 - 4z^2}}{2z}$$

where the negative sign is chosen so that $\tilde{G}_A$ does not blow up at 0 (by L'Hopital's rule). A consequence of this is that:

$$G_A(z) = \frac{z - \sqrt{z^2 - 4}}{2}$$

which *does* decay as $z \to \infty$ (Why?).

Now we want to find the functional inverse by solving:

$$\begin{aligned} w &= G_A(z) = \frac{z - \sqrt{z^2 - 4}}{2} \qquad \Rightarrow \qquad (2w - z)^2 = z^2 - 4 \\ &\Rightarrow \qquad -4wz + 4w^2 = -4 \qquad \Rightarrow \qquad z = \frac{1}{w} + w, \end{aligned}$$

i.e. we have

$$G_A^{-1}(w) = \frac{1}{w} + w.$$

Thus the R-transform is particularly simple:

$$R_A(w) = G_A^{-1}(w) - \frac{1}{w} = w$$

which is also the R-transform for B. Thus we know

$$R_{A+B}(w) = 2w \qquad \Rightarrow \qquad G_{A+B}^{-1}(w) = \frac{1}{w} + 2w.$$

We can now solve

$$z = \frac{1}{w} + 2w \qquad \Rightarrow \qquad zw = 1 + 2w^2 \qquad \Rightarrow \qquad w = \frac{z - \sqrt{z^2 - 8}}{4}$$

i.e. we know that

$$G_{A+B}(z) = \frac{z - \sqrt{z^2 - 8}}{4}.$$

We leave the final step of finding the moments and distribution as a possible poster project. But actually from this formula we can do an educated guess: since the square-root vanishes at $z = \pm\sqrt{8} = \pm 2\sqrt{2}$ we can guess that the distribution is a semicircle with the same endpoints, i.e.,

$$\frac{\sqrt{8 - x^2}}{4\pi}.$$

III.4 Poster projects

Here are some ideas for poster projects related to free probability:

1. Investigate why two Wigner ensembles A_n and B_n follow the rules of free probability as $n \to \infty$.
2. Understanding all free probability operations including multiplication and truncation (e.g., the limit of $A_n[1 : n - m, 1 : n - m]$). These operations can also be reduced to simple functions of the input moments, and one could walk through low orders.
3. Explore the Free Central Limit Theorem either mathematically or experimentally, where $A_1 + \cdots + A_n$ tends to a semicircle when A_k are free and independent.
4. Complete the derivation of the addition of two semicircles and explore further examples like the Marchenko–Pastur distribution.
5. Do additions and multiplications of random matrices still exhibit universality in local statistics?

Chapter IV

Randomised Linear Algebra

Our last topic is *randomised linear algebra*: how can we incorporate random matrices into developing better algorithms for linear algebra? Numerical linear algebra (that is to say, linear algebra done on computer using algorithms) underlies much of modern computational mathematics, including the numerical solution of Partial Differential Equations (PDEs). Recently, incorporating randomness into numerical linear algebra has become a tool to design algorithms that can scale to much larger problems than non-randomised algorithms. For example, Stochastic Gradient Descent is a randomised version of a linear algebra algorithm (Gradient Descent, or in the linear case Conjugate Gradients) that underlies training of Neural Networks in Machine Learning. Without randomness, training can get stuck in local minima, and so it is absolutely essential.

In this section we will focus on a much simpler problem which can in many ways be viewed as a baby version of training a Neural Network: linear regression. (Indeed, linear regression can be thought of as a ”one-layer” Neural Network.) We will first review least squares, where rectangular systems are “solved” by finding a vector that minimises the error in norm, with linear regression being a classic application. We then explore experimentally how using a random *sketch matrix* can substantially reduce the sizes of matrices involved, and by averaging over multiple sketches we can obtain a good approximation to the true least squares system.

Further Reading For simplicity we focus on experimental implementation and observation of randomised linear algebra, but Martinsson & Tropp 2020 gives an overview of the mathematical proofs of what we observe. It also discusses some other simple problems that could form good posters like trace estimation (approximating the trace of a matrix without evaluating every single diagonal). Randomised linear algebra is a very active research area and so there will probably be a lot of progress in the next few years.

IV.1 Least squares

We begin with an introduction to least squares. For a rectangular matrix $A \in \mathbb{R}^{m\times n}$ with $m > n$ and a right-hand side $\boldsymbol{b} \in \mathbb{R}^m$ we cannot solve $A\boldsymbol{c} = \boldsymbol{b}$ unless $\boldsymbol{b}$ happens to be in the column span of A. However, we can instead find $\boldsymbol{c} \in \mathbb{R}^n$ that satisfies $A\boldsymbol{c} \approx \boldsymbol{b}$, in the sense that

$$\|A\boldsymbol{c} - \boldsymbol{b}\|$$

is minimised. This is called the *least squares* solution. Minimising a norm is equivalent to minimising its square, and expanding out we see that it is equivalent to a quadratic minimisation problem:

$$\|A\boldsymbol{c} - \boldsymbol{b}\|^2 = (A\boldsymbol{c} - \boldsymbol{b})^\top (A\boldsymbol{c} - \boldsymbol{b}) = \boldsymbol{c}^\top A^\top A\boldsymbol{c} - 2\boldsymbol{c}^\top A^\top \boldsymbol{b} + \boldsymbol{b}^\top \boldsymbol{b}.$$

In the scalar case ($n = 1$ so that $\boldsymbol{c}$ is essentially a constant) we know we can solve this by differentiation with respect to $\boldsymbol{c}$, and we arrive at a linear system

$$A^\top A\boldsymbol{c} = A^\top \boldsymbol{b}.$$

This generalises (actually, using linear algebra instead of calculus but we omit the details) and in general we know the least squares solution is given by

$$\boldsymbol{c} = (A^\top A)^{-1} A^\top \boldsymbol{b}.$$

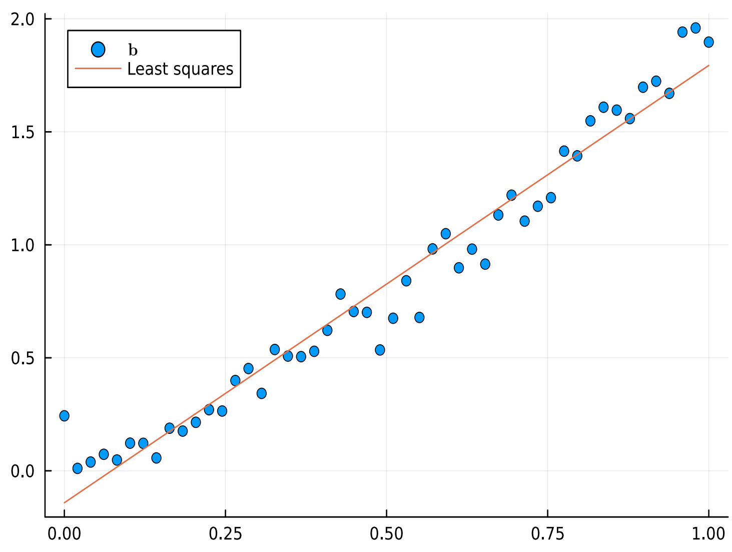
Assuming A has full column rank we know $A^\top A \in \mathbb{R}^{n\times n}$ is invertible (Why?) and so this is well-posed.

A classic application of least squares is *polynomial regression*. In the case of linear regression, we want to approximation data $\{b_j\}$ on a grid of points $\{x_j\}$ by a linear function $c_0 + c_1 x$. In other words, we want to solve

$$\underbrace{\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} c_0 \\ c_1 \end{bmatrix}}_{\boldsymbol{c}} \approx \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}}_{\boldsymbol{b}}$$

by using the least squares solution.

Here is a depiction of the linear regression fit of (noisy) data at $n = 50$ points:

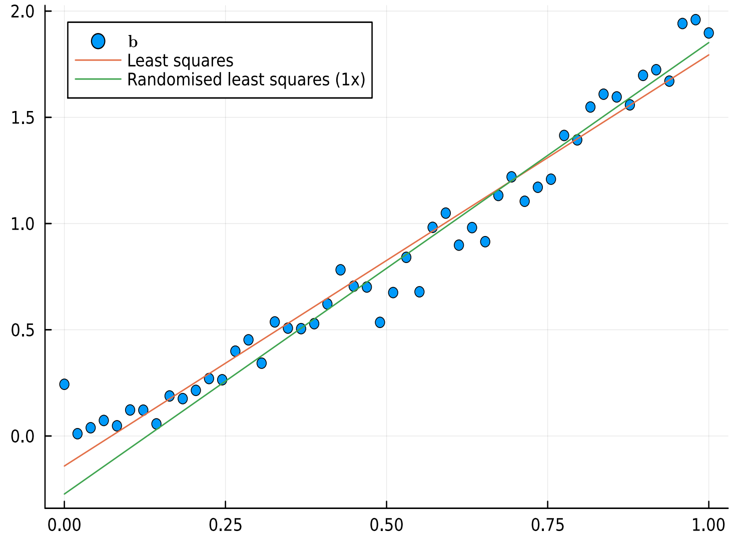


IV.2 Randomised least squares

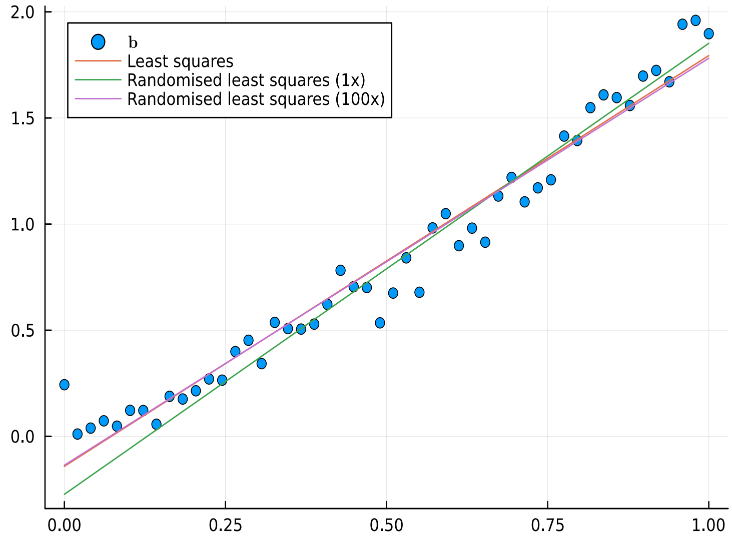
In the above data there is a lot of correlation between nearby points, hence, using every single point is unnecessary. An alternative is to *sketch* the problem: use a sketch matrix $P \in \mathbb{R}^{r\times m}$ where $r \ll m$ and solve

$$PA\boldsymbol{c} \approx P\boldsymbol{b}$$

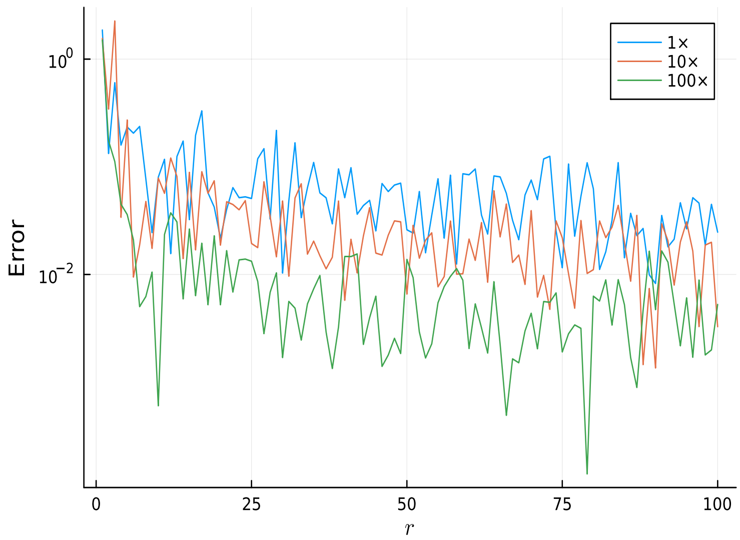
using least squares. Here we see a simple example where we use $P \in \mathbb{R}^{5\times 50}$ with random entries:



This is a single sample and, to the eye, it performs fairly well even compared to the true least squares solution. Going further, we see that averaging over multiple trials is almost indistinguishable from the true least squares approximation:



Now this particular example is a bit silly: its not at all clear that solving 100 least squares systems of size 5×2 is preferred to a single least squares system of size 50×2. But we can consider a much more extreme example of $n = 10,000$. Here we compare the average over samples for varying r:



That is to say, we can achieve accuracy to within about 1% by solving 100 least squares problems involving 10×2 matrices. I claim this is substantially faster, and so if 1% accuracy suffices (which it does in *many* applications) it is a big improvement.

IV.3 Poster projects

We will leave further investigation of randomised linear algebra to posters:

1. How does the approximation property depend on the distribution of the sketch matrix? Can we use different sketch matrices to get better accuracy?
2. What other algorithms are amenable to randomisation (trace estimation, low rank approximation, norm estimation, multiplication)?
3. What applications are there to randomised linear algebra such as higher order polynomial regression?